

Ritabrata Chakraborty

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EDUCATION

University of Pennsylvania

M.S.E. in Robotics, GPA: -/4

Pennsylvania, USA

Aug '26 – May '28

Birla Institute of Technology and Science, Pilani (BITS Pilani)

Rajasthan, India

B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10

Oct '22 – May '26

- **Institute Merit-cum-Need Scholarship** — Consecutive Recipient (Top 8% of Students)

PUBLICATIONS & PEER REVIEWS

Journal Publications

- R. Chakraborty and K. Kishore, “*Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.*” Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026. [🔗](#)
- R. Chakraborty and K. Kishore, “*Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.*” Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, R. Chakraborty, and G. Sartoretti, “*CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.*” Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**. [🔗](#)
- R. Chakraborty, T. Mian, and P. Kundu, “*An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.*” Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**. [🔗](#)
- K. Kishore and R. Chakraborty, “*Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.*” Publication awaiting intellectual property clearance, 2025. [🔗](#)

Peer Reviews

- “*GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.*” Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- “*Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.

EXPERIENCE – ROBOTICS & COMPUTER VISION

Multi-Critic Decomposition for Autonomous Indoor Navigation [🔗](#)

Supervisor: **Dr. Kaushal Kishore**, Principal Scientist, CSIR-CEERI

Jan '26 – May '26

- Integrated three specialized critics for forward progress, lateral avoidance, and angular alignment with distinct reward signals.
- Deployed multi-critic TD3 with prioritized replay achieving 94.2% success rate and 22% lower collision rate versus baseline.

ML Intern | Uber (AI Solutions)

Supervisor: **Mr. Siddarth Malreddy**, Tech Lead Manager & **Mr. Ishan Nigam**, Senior ML Engineer, Uber

Jul '25 – Dec '25

- Augmented **uLabel** with object tracking for automated RGB/IR video annotation, reducing manual effort.
- Applied XGBoost anomaly detection (mAP50: 75%) to flag perturbed bboxes in human-in-the-loop tracking validation.

Design and Development of Smart Automated Field Hockey Ball Launcher [🔗](#)

Supervisor: **Dr. Mani Shankar Dasgupta**, Associate Professor, BITS Pilani

Apr '24 – Oct '25

Physics-Based Launcher Mechanical Design and FEM Analysis

- Fabricated programmable launcher achieving 150 km/h using CFRP composite and stainless steel components.
- Performed FEM analysis and CAD using Fusion 360, achieving safety factor 2.07 at 496 RPM.

Computer Vision Pipeline for 3D Trajectory Reconstruction and Shot Classification

- Devised 3D ball localization from monocular videos using diameter-based depth and **PnLCalib** (0.76m RMSE).
- Constructed TCN-attention classifier fusing trajectory and statistical features, achieving 96.4% classification accuracy.

Uncertainty-Guided Path Planning via Conditional Layout Prediction [🔗](#)

Supervisor: **Dr. Guillaume Sartoretti**, Assistant Professor, MARMoT Lab, NUS

Mar '25 – Aug '25

- Adapted multiple image-based exploration planners into ROS pipelines for evaluation in the **CMU Exploration Environment**.
- Supported the development and benchmarking of **CogniPlan**, achieving up to 17.7% shorter exploration paths and 3.9% improved navigation efficiency over state-of-the-art baselines across 100+ unseen maps.

Monocular Vision-Based UAV Navigation for Orchard Monitoring [🔗](#)

Supervisor: **Dr. Kaushal Kishore**, Principal Scientist, CSIR-CEERI

Jan '24 – Feb '25

- Engineered a UAV-based orchard monitoring system using YOLOv11 (Box mAP50: 95.5%, Mask mAP50: 96.5%).
- Implemented B-spline trajectory logic and coded custom yaw-roll controller to minimize drift under wind.

EXPERIENCE – GENERATIVE AI & PREDICTIVE MAINTENANCE

Synthetic Data Generation for Bearing Fault Diagnosis Under Data Scarcity <i>Supervisor: Dr. Pradeep Kundu, Assistant Professor, KU Leuven</i>	<i>Sep '24 – May '26</i>
Auxiliary Classifier WGAN-GP for Time-Series Sensor Data Generation ↗ - Built ACWGAN-GP with TCN discriminator for minority fault class augmentation reaching ~100% classification accuracy. - Assessed synthetic data using PCC, Cosine Similarity, MMD, and KL Divergence on FFT spectra.	
Physics-Informed Generative Modeling for Bearing Fault Diagnosis Under Data Scarcity - Formulated physics-informed cVAE-GAN and cVQ-GAN frameworks isolating latent-space effects in CWT scalogram synthesis. - Validated LiteFormer classification gains across Density, Coverage, and CMMD metrics.	

PROJECTS

Autonomous Drone Navigation MathWorks Global Student Drone Challenge 2025	<i>Mar '25 – Apr '25</i>
- Designed vision-based control for Parrot Mambo drone using masking, ray-tracing, and closed-loop yaw control. - Optimized speed control and zone-based auto-landing to reduce track completion time.	
ExoMy Rover Navigation and UR3 Arm Motion Planning ERC 2023 Remote ↗	<i>Apr '23 – Sep '23</i>
- Navigated ExoMy rover using ArUco detection, Ackermann steering, and spot turns for autonomous hazard avoidance. - Calibrated UR3 arm with MoveIt and OMPL planner for collision-free manipulation (98% success).	

AWARDS & ACHIEVEMENTS

Finalist — Meta PyTorch OpenEnv Hackathon (India) ↗	<i>Apr '26</i>
3rd Place — MathWorks Global Drone Student Challenge 2025	<i>Mar '25</i>
Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) ↗	<i>Jul '24</i>
Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 ↗	<i>Jul '24</i>
5th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)	<i>Sep '23</i>

TEACHING

Teaching Assistant <i>CS F320: Foundations of Data Science, BITS Pilani</i>	<i>Jan '26 – May '26</i>
<i>ME F218: Advanced Mechanics of Solids, BITS Pilani</i>	<i>Jan '25 – May '25</i>
<i>ME F216: Materials Science and Engineering, BITS Pilani</i>	<i>Sep '24 – Dec '24</i>
- Assisted 100+ students in labs and tutorials, clarifying concepts and linking theory to practical applications. - Analyzed assignments and facilitated faculty in delivering high-impact teaching sessions.	

LEADERSHIP & VOLUNTEERING

Deputy Director of Academic Programming <i>Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania</i>	<i>Aug '26 – Present</i>
Vice-President <i>South Asian Association (Rangoli), University of Pennsylvania</i>	<i>Aug '26 – Present</i>
President & Secretary <i>Mechanical Engineering Association (MEA), BITS Pilani</i>	<i>Jun '24 – May '26</i>
- Coordinated 10+ events and career sessions for 300+ students, facilitating technical exposure and alumni interaction. - Managed production of 500+ merchandise items and led outreach, boosting student participation by 20%.	
President & Tech Fest Coordinator <i>Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani</i>	<i>Oct '24 – Jul '25</i>
- Led a team of 25+ members to organize 4+ technical workshops with HVAC industry experts. - Hosted 3 competitions and networking events, engaging 200+ students in HVAC innovation and awareness.	
Project Manager <i>Tinkerer's Lab (TL), BITS Pilani</i>	<i>May '24 – Jul '25</i>
- Supervised 5 interdisciplinary robotics teams (30+ members) on Micromouse and Hexapod projects. - Oversaw lab resources, conducted weekly reviews, and mentored 50+ students in hands-on technical skills.	

TECHNICAL SKILLS & COURSEWORK

Relevant Coursework: Machine Learning, Deep Learning, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Computer Programming, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins

Programming Languages: Python, C++, C, Shell (Linux)

Robotics & Simulation: ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, OpenCV, Open3D, Matplotlib, Weights & Biases (W&B)

Hardware & Embedded Systems: NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR

CAD & Mechanical Simulation: ANSYS Mechanical, SolidWorks, Fusion 360